

Exploring the project manager and project management in agricultural operations: a review

Caio Senna do Amaral* and
Fernando Tobal Berssaneti

Production Engineering Department,
University of Sao Paulo,
Av. Prof. Luciano Gualberto – 1380, Butantã – 05508-010,
São Paulo/SP, Brazil
Email: amaral.caio@usp.br
Email: fernando.berssaneti@usp.br
*Corresponding author

Abstract: This article analyses how the project manager and project management methodologies are used in the agricultural supply chain. The methodological approach was a systematic literature review based on bibliometric techniques, network analysis, and content analysis. The literature analysis reveals that despite the constant use of projects as a way of obtaining strategic results for the agricultural supply chain and the use of project management techniques and tools by this community, there is no mention of the role of the project manager and very little of the use of methodologies showing a low level of maturity in terms of project management. As a practical application, this paper exposes the need to discuss traditional models of the structure of the agricultural supply chain, suggesting the inclusion of the project manager as an agent of business integration. The originality of this paper is reviewing the role of both project management and the project manager in the agricultural market.

Keywords: project manager; project management; agriculture; supply chain.

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Biographical notes: Caio Senna do Amaral is a Master's candidate at the University of São Paulo (USP) on the Production Engineering Department of the Polytechnic School in Brazil. He is a member of the Quality and Product Engineering Research Group. He holds a BSc in Industrial Chemist from the Federal University of São Paulo (UNIFESP) and his current research focuses on project management, agricultural business and quality management.

Fernando Tobal Berssaneti holds a Degree in Production Engineering from the Polytechnic School of the University of São Paulo, a Master's in Production Engineering from the Polytechnic School of the University of São Paulo in 2006 and a PhD in Production Engineering from the Polytechnic School of the University of São Paulo in 2011. He is Professor of the Production Engineering Department of the Polytechnic School of the University of São Paulo and member of the Board of Trustees of the Carlos Alberto Vanzolini Foundation. He works as a business consultant and teaches specialisation courses at Escola

Politécnica da USP since 2004. He has experience in the area of production engineering, with emphasis on economic engineering, financial mathematics, project and product management and quality and productivity management.

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1 Introduction

1.1 *Agricultural supply chain and operations*

The increase in the global population has challenged the demand for agricultural and food products. This challenge includes delivering goods and services considering the ecological environment, public health, and animal welfare (Hvitsand and Leikvoll, 2023).

Thereby, the production of various items in the agricultural food supply chain has increased to mitigate food security concerns. On the other hand, the excessive production of products has led to various issues due to a complex business-to-business-to-customer market model that makes agricultural supply different from other markets (Gholian-Jouybari et al., 2023).

Agriculture supply chain has an essential difference from the others supply chain. The production depends on the farmers’ practices and demands for knowledge of the proper food chain. Another critical topic is animals or humans can consume those final goods (Kumari et al., 2022).

In addition, these factors must consider the need for industrialisation of the sector, the need for more accuracy of data and decision-making, the inefficiency of the supply chain, and a delay in applying management techniques (Kamble et al., 2020; Yücenur et al., 2022).

The agricultural supply chain is a highly complex market based on monoculture and intensive animal farming. It is usually divided into five stages, from farm to customers. These steps are namely

- 1 food production
- 2 industrial processing
- 3 distribution
- 4 retail
- 5 consumption.

After the production by farmers, these products are sent to industrial processing. The products are then aggregated to the wholesalers in each region through distribution services. So, then food retailers in the region need to place orders for the food products from the wholesalers to make food products available for sale. Finally, consumers can purchase and consume the goods (Liu et al., 2021).

According to the Food and Agriculture Organization (FAO) of the United Nations, more than 30% of the world's food produced yearly is not consumed due to food loss and food waste. Food loss occurs at every stage of the agricultural supply chain, especially during food harvesting, sorting, grading, and processing. Some lessons learned in the agricultural supply chain were already discussed in the literature showing a gap compared to the traditional supply chain. Topics such as food safety issues, asymmetric and fragmented information, and work management (Amaral and Berssaneti, 2022), and an inefficient supply chain are recurrent (Liu et al., 2021).

1.2 Project manager and project management in agribusiness

The operational complexity of the agricultural supply chain is an operational and management challenge for today's world. Factors such as seasonality of production, extended and pre-determined production periods, variety of quantity and pattern specifications, safety stock restrictions, and traceability expose this chain to supply dysfunctions (Nyamah et al., 2017; Thume et al., 2022).

Project management has already established itself as one of the tools for developing a more strategic vision within organisations. Different authors exposed the need to use project management in agriculture to implement emerging Technologies and process improvement (Łukasz et al., 2022) (e.g., internet of things, robotics, artificial intelligence, big data analytics, and blockchain), with the primary objective of making the supply chain more reliable and exposing food to less risk (Liu et al., 2021).

It has already been widely discussed in the literature that the role of the project manager and project management techniques are considered pivotal in projects (Shastri et al., 2021) involving multiple aspects of managing teams such as leadership, team building, motivation, communication barriers, influencing, decision making, planning, and coaching (Alves et al., 2022).

Organisations that have a PM role and PMO structure well defined and in place for aligning the project deliverables with their organisational goals will be better (Sergeeva and Ali, 2020; Rotimi et al., 2022) placed to realise their investment in projects and achieve the value defined by their business strategies (Too and Weaver, 2014). However, its role and structure in the agricultural market remain unclear (Garcia and Jerez, 2020). The institution of project management methodology involves cultural changes, skills acquisition, and technical and managerial tools (Carvalho et al., 2018). These support the resolution of complex problems such as sustainable land management, climate change, rural resilience, and the management of typical agribusiness stakeholders.

In this sense, it is essential to understand the inherent characteristics of the agricultural supply chain and how the project manager and project management methodologies can adapt to ensure its solid and sustainable growth, in addition to acting as business integrators in the agricultural supply chain in agriculture.

This article aims to answer the following research question (QP):

QP What is the project manager's role, and what are project management techniques in the agricultural supply chain?

In order to answer the research question, an in-depth literature review was conducted, applying bibliometric and content analysis with different software. This paper identifies correlations between project management, the project manager, and supply chain

operations in the agricultural business field. For the agribusiness community, the study shows the critical role of project management in solving complex problems.

The article is structured into five sections. Next, a synthesis of the theoretical discussion of the literature and fundamental conceptual definitions are presented (Section 2), as the methodological approach used in the research (Section 3), results (Section 4), and conclusions (Section 5).

2 Literature review

2.1 *Agriculture as a way of obtaining a product, service or result*

In 2007, Rao highlighted that agricultural production methods would determine how food production will be structured and how the environment will be organised. It is in this context that we must understand the agricultural supply chain and the use of project management techniques. Although the topic has been extensively studied in other business areas (Carvajal et al., 2019), there still needs to be more material in the literature that presents the integration of the agricultural supply chain.

The agricultural supply chain involves several activities: planting, development, harvesting, storage (on the farm and in warehouses), processing, transport, and distribution. As an extra complicating factor for these activities, connections in material flow, weather conditions, yield variations, and long production cycles must also be considered (Carvajal et al., 2019).

It is common to try to compare the use of tools, techniques, and strategies between industry and agriculture. However, the conclusions derived from this comparison must consider two basic rules: the use of equivalent technologies or efforts and the use of equivalent methodologies (Grammig, 2003).

However, there is a similarity with the supply chain of other markets regarding performance measurement in project management. Nyamah et al. (2017) used the concept of the total cost, customer response, flexibility, time, process, and food quality and safety as criteria for measuring agricultural chain performance.

Nevertheless, technological advances in the production process demand practitioners' knowledge and project management experience (Nagy et al., 2009). Among different sectors of industry or services, researchers have already demonstrated the potential of management systems to generate sustainable performance in their market (Darian et al., 2018).

Overall, the literature has highlighted reducing food waste and increasing efficiency as crucial elements to drive sustainability in the agricultural supply chain. Hence, systematic project management involves applying structured processes using methodologies, tools, and models to standardise best practices (Ibert, 2004).

2.2 *Dynamic competencies and the project manager as a role of business integration*

Projects are used as tools to support organisations in strategic restructuring due to changes in the market. Several studies indicate that project success is closely related to the manager's competencies (do Vale et al., 2018).

Firstly, Verzuh (2005) compares the project manager to an orchestra conductor since he harmoniously unites several groups or parts of the project. For this, it is necessary to possess or present characteristics that can be technical or behavioural (Gruden and Stare, 2018).

Furthermore, emotional skills may be required as a project is a temporary endeavour. Generally, dynamic competencies are associated with the ability to adapt according to external changes in order to obtain business advantages. According to Helfat et al. (2007), dynamic competencies mean skills to maintain, create and extend or modify an organisation through management improvement (Gulino et al., 2020).

In addition, Güngör and Gözlü (2017) consider knowledge management in companies oriented to projects at a strategic level, which is an important way of creating a memory and deriving lessons from experience.

Raymond et al. (2016) add that knowledge management needs accessible and quality knowledge assets and effective mechanisms to share this knowledge among project managers. Another point to be considered is the Project Management Office (PMO), which plays a prominent role in the management and dissemination of knowledge (Aubry et al., 2011). More than over, the exchange of knowledge between partners tends to be reciprocal to generate benefits for the project (Alves and de Carvalho, 2022).

From this, it is necessary to understand how the integration of knowledge from different areas occurs. Knowledge integration is a process in which competitive advantage is gained by complementing existing knowledge from different individuals. This type of action is necessary when knowledge is specialised and dispersed. Two methods are generally used: the first through frequent communication and intense sharing, and the second depends on the structural mechanisms of the organisation. Integrating knowledge forces individuals to “communicate, confront and anticipate concerns, attitudes and perceptions” (Enberg, 2012).

3 Materials and methods

The methodologies considered in answering the research question were a systematic literature review, bibliometric analysis, network analysis, and content analysis.

A systematic literature review was used in this article because it is a methodology capable of enabling the analysis of specific topics with the necessary rigor and transparency. This methodology aims to find scientific material based on pre-established criteria that answer this article’s research question. In this way, through systematic methods, conclusions can be drawn (Snyder, 2019).

This study also carried out a bibliometric analysis, network, and content analyses to find the main scientific works and relevant authors and the main topics, approaches, and methods. When properly categorised and analysed, bibliometric analysis allows authors to define their constructs with other scientists’ data, enabling new contributions to the literature (Zupic and Carter, 2015).

In addition, these methods are complementary and help identify the prominent authors and publications, thus enabling a better clarity of the existing literature (Zupic and Carter, 2015). These methods are justified by allowing an evolutionary analysis of project management tools and methodologies in the agricultural supply chain over the years. Thus, in this way, there is a correlation between the different project management techniques and the evolution of agriculture, presenting an evolutionary analysis.

3.1 Sampling process/sampling process

The initial sample was constructed using the scientific reference base ISI Web of Science, which includes publications from other databases (Scopus, ProQuest, and Wiley). It allows a critical bibliometric analysis with different types of information, such as the number of citations, institutions, countries, and impact factors. All articles published up until March 2023 were considered.

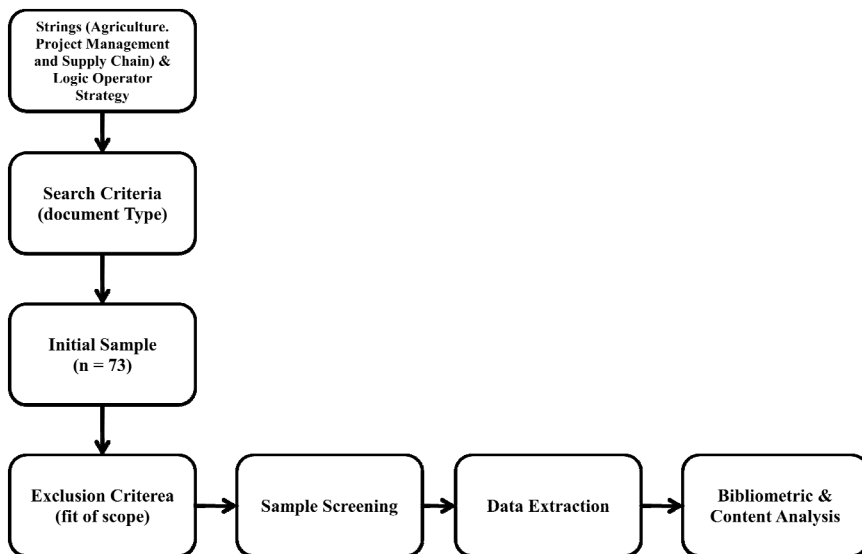
The keywords used for the search were ‘agriculture’, ‘project management’, and ‘supply chain’, and the Boolean operator used was ‘and’. Thus, the search allowed us to find publications that contained the intersection of the three keywords.

The initial result of the search was 73 publications, among which were included: articles, meetings, reviews, proceedings papers, and patents. Only articles, proceedings papers, and reviews were considered for this study.

These publications were accepted by 60 journals, classified into ten research areas, and published between 2001 (first published article) and 2023 (last published article). The base was considered for the bibliometric analysis and for the network analyses.

For content and coding analysis, a second screening was performed. After reading the abstracts of the articles, 31 articles were excluded. The exclusion was performed because, even if the article mentioned one of the keywords, its content was not directly related to the study topic. Another three articles were included in the ‘snowballing’ process. Figure 1 outlines how the activity of choosing articles was carried out.

Figure 1 Research flow



3.2 Data analysis

The data analysis had as its primary objective the answering the research question proposed by this paper. Therefore, the bibliometric analysis, network analysis, and

content analysis were directed toward a better understanding of the relationship between project management in the agricultural supply chain.

Network analyses are graphical analyses that allow us to understand the relationships of centrality and intensity between authors, keywords, and references. With this in mind, it is possible to understand which are the prominent authors who work at the intersection of search keywords ('agriculture', 'project management', and 'supply chain') as well as the words most commonly used by these authors in their works (Chen, 2004).

It is worth noting that this paper sought to understand which other words, in addition to the keywords, are present in the bibliography found and also, through the analysis of the references, to list which are the leading publications that serve as a theoretical reference for the researchers.

Additionally, the following analyses were carried out: an overview of the evolution of publications on the subject, graphical and discrete statistical analyses, and content analysis.

Content analysis is a measurement method that has gained popularity in management because it has advantages for identifying and summarising trends and measures of the constructs proposed by researchers. This type of analysis has many methodological advantages compared to other methods. The possibility of managing a large volume of disordered data and reducing researchers' biases during data analysis stands out. A well-executed content analysis follows the following steps: data collection, coding analysis, and interpretation of the coded content (Gaur and Kumar, 2017).

3.2.1 Bibliometrics

Minitab software was used for the studies of bibliographic evolution, graphical analysis, and discrete statistics based on the metadata extracted from the ISI Web of Science reference base described in Subsection 3.1.

3.2.2 Network analysis

Network analysis was performed using the VOSviewer software version 1.6.15 for the Microsoft Windows operating system.

3.2.3 Content analysis

Content analyses were performed using 28 articles in the reference base. For this, the research protocol proposed by Tranfiled et al. (2003) and Duriau et al. (2007) was used, with the following steps:

- 1 planning the review – defining the research question, search strategy, and coding
- 2 conduct the review – counting frequency and cross-tabulation
- 3 reporting and dissemination (interpretation of results).

NVivo software was used in the sample encoding process. For the coding of the articles, the following categories were used: type of study, affiliation, financial support, period of analysis, approach, geographic scope, unit of analysis, and source of evidence. Table 1 presents the results found.

In order to find the terms that occur most frequently in the database, the NVivo software was used to determine the relative frequency percentage. The exact correspondence of the words was also defined as a grouping criterion. Words of the exact origin (such as gerunds and participles), synonyms, specialisations, and generalisations were not considered.

Thus, the 50 most frequent words that met the above criteria represented 12.69% of the weighted percentage, representing 4,804 appearances. The most frequent words were model (0.68%), supply (0.61%), business (0.58%), chain (0.48%), and production (0.43%).

Table 1 Code analysis

<i>Category</i>	<i>Subcategory</i>	<i>Frequency (%)</i>
Methodological approach	Theoretical-conceptual	21
Methodological approach	Simulation	29
Methodological approach	Literature revision	11
Methodological approach	Survey	7
Methodological approach	Case study	14
Methodological approach	Research	18
Filiation	University	85
Filiation	Research institution	11
Filiation	Company	4
Financial support	Yes	39
Financial support	No	61
Analysis period	Hindsight	64
Analysis period	Contemporary	32
Analysis period	Longitudinal	4
Approach	Quantitative	39
Approach	Qualitative	54
Approach	Descriptive	4
Approach	Predictive	4
geographic scope	National	89
geographic scope	International	11
PM methodology	No	100
PMO structure	No	100
Project manager	No	100

4 Results

4.1 Overview of literature: evolution, main journals, authors and topics

From the database, a series of analyses were carried out to present an overview of the existing literature. The complete database was considered for these analyses, with the total number of publications.

It is possible to see a 400% growth in the number of publications in 2018 compared to 2017 and a 200% growth in 2019 compared to 2018, indicating an increase in the academic community's interest in research.

At the same time, it is essential to realise that there are few publications per year, which allows us to interpret that the research area is recent, a field where new contributions can be made.

Discrete statistical tools were used to analyse citations in the search terms defined in this research. This type of information allows us to understand these publications' bibliometric significance. Table 2 shows that the average number of citations of the 28 analysed publications is 26.2 citations per article, a relatively low number compared to isolated search terms when the intersection of terms is disregarded. The average standard deviation of 67.4 confirms the low homogeneity of the sample, which presents minimum citations as 0 and a maximum of 354.

Table 2 Citation analysis

Variable	N	N*	Mean	DevPad	Minimum	Median	Maximum
Citation	28	0	26.2	67.4	0	8.5	354

Collaborating for a deeper analysis of the total citations allows us to notice the highlights with citations above 45. Carlsson and Rönnqvist (2005) published these articles (Rao, 2007; Zhang et al., 2016; Ahumada and Villalobos, 2009).

Significant numerical contributions were in agriculture, multidisciplinary, computer science, interdisciplinary applications (n = 6.24%), business; management (n = 2.8%).

The presence of the terms agriculture, business, and management suggests that the intersection of the terms of this research results in a strategic interest of companies in associating project management with financial and business results.

An analysis of the leading journals used by the research groups that composed the database of this paper was also carried out, and the result is shown in Table 2.

Computers and Electronics in Agriculture have the highest number of citations (n = 6), followed by the *European Journal of Operational Research* (n = 2). All other journals contained in the database had only one publication, of which the following stand out by the number of citations: *European Journal of Operational Research*, *Computers and Electronics in Agriculture*, *Technological Forecasting and Social Change*, *Renewable Energy*, *International Journal of Advanced Manufacturing Technology*, and *Business Strategy and the Environment*.

The impact calculated by the journal citation report was used as a quality reference for the journals in the database. The following journals stand out: *European 257 Journal of Operational Research* (JCR = 4,213), *Computers and Electronics in Agriculture* (JCR = 3,858), *Technological Forecasting and Social Change* (JCR = 5,846), *Renewable Energy* (JCR = 6,274), *International Journal of Advanced Manufacturing Technology* (JCR = 2,633) and *Business Strategy and the Environment* (JCR = 5,483).

4.2 Network analysis

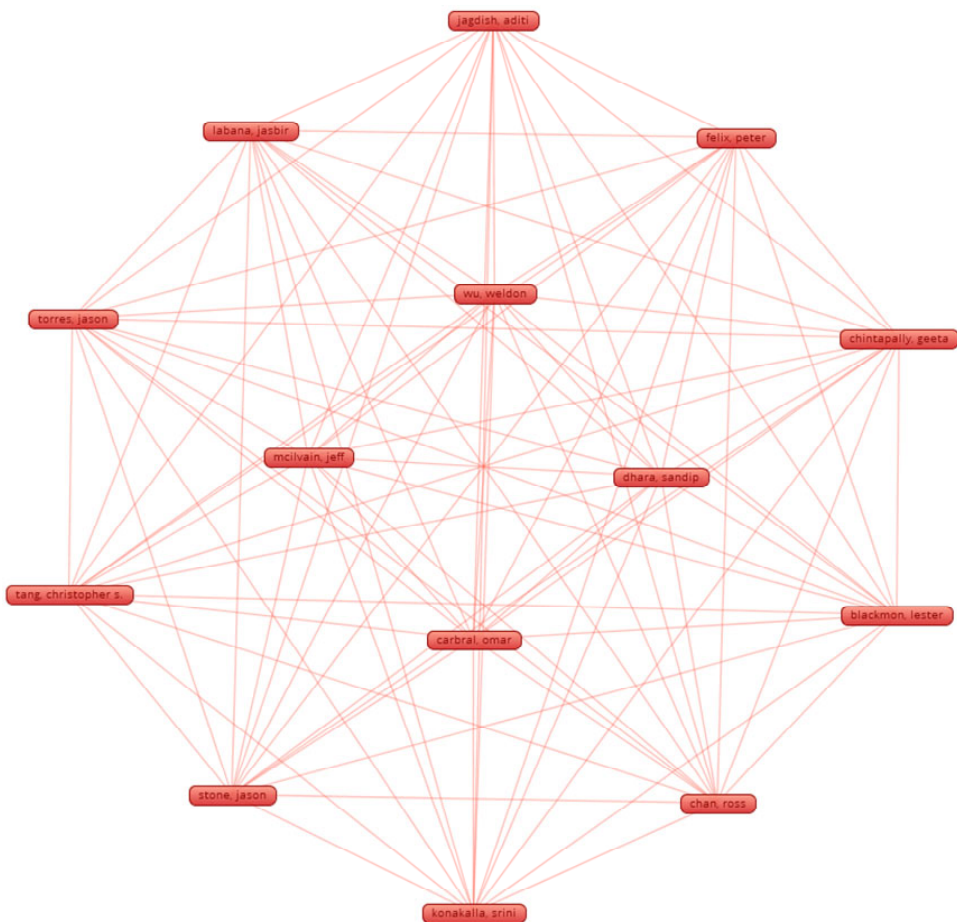
The result of the network analyses (co-authorship, co-occurrence, and co-citation) carried out in this paper are presented in the sections below.

4.2.1 Co-authorship analysis

The minimum number of documents and citations per author for the co-authorship analysis was 1. In this network analysis, the author who presents at least one citation in one document would appear in the network analysis.

The database had 286 authors, and the number of authors was maintained after the above-mentioned definitions. All 268 authors were chosen to form the network, but only 14 authors connected. Figure 2 indicates the presence of one large cluster. It is important to note that all authors connect with all authors.

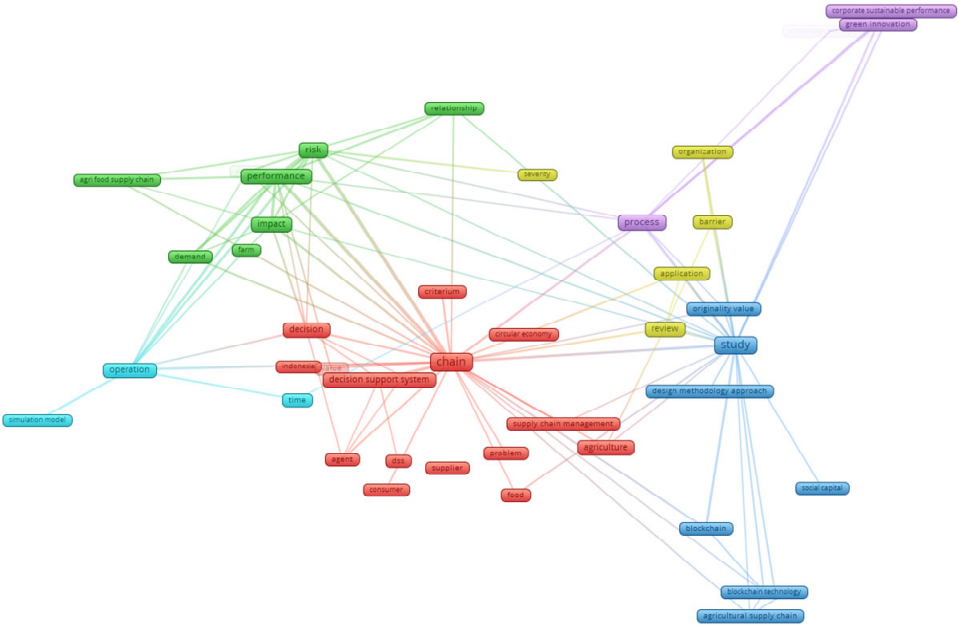
Figure 2 Co-author analysis (see online version for colours)



4.2.2 Co-occurrence analysis

For the co-occurrence analysis, all keywords were defined as the study unit. The minimum number of occurrences was defined as seven words, which means that for the network map generation, the keywords should be connected with at least seven other words. The final analysis network contained 44 words that formed 6 clusters (Figure 3).

Figure 3 Concurrency analysis (see online version for colours)



Six clusters were generated by grouping according to the words shown in Table 3. These clusters indicate the strong path between traditional operations management practices without a project manager or a Project Management Office.

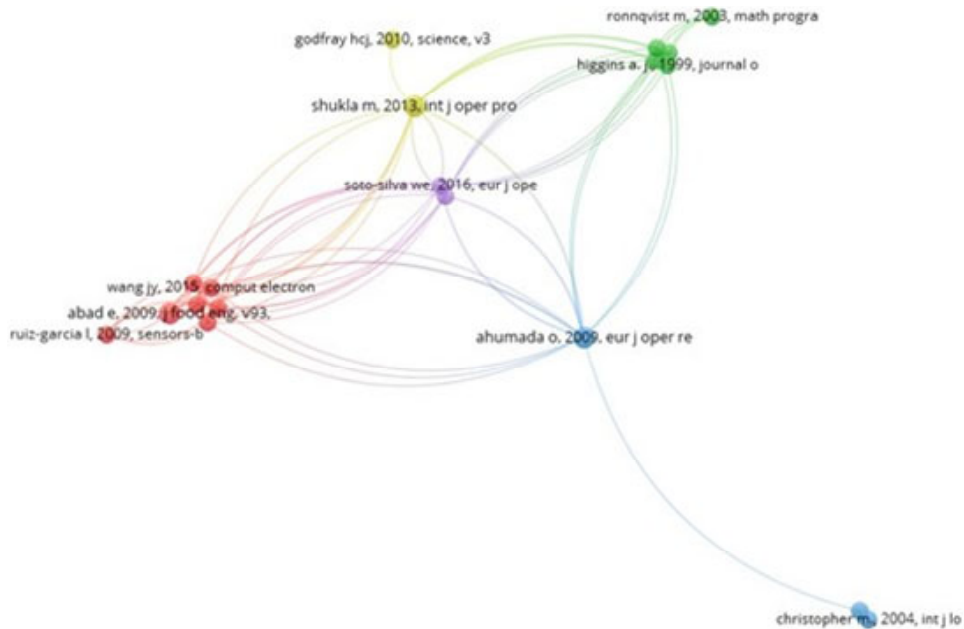
Table 3 Concurrency analysis

Clusters	Word
Cluster 1	Agent, agriculture, chain, circular economy, customer, criterium, decision, decision support system, DSS, food, Indonesia, order, problem, supplier, supply chain management, and value
Cluster 2	Agrifood supply chain, demand, farm, impact, performance, relationship, risk, and supply
Cluster 3	Agriculture supply chain, blockchain, blockchain technology, COVID, design methodology approach, originality value, social capital, and study
Cluster 4	Application, barrier, organisation, review, and severity
Cluster 5	Corporate sustainable performance, green innovation, knowledge management, and process
Cluster 6	Operation, simulation model, and time

4.2.3 Co-citation analysis

In the citation analysis, the study unit was defined as authors, and the minimum number of author citations was defined as 2. Each author should connect with at least two more authors. Five clusters were generated, as described in Figure 4. Three authors stand out in this article's research area (Ahumada and Villalobos, 2009; Shukla and Jharkharia, 2013; Abad et al., 2009).

Figure 4 Analysis of co-citation (see online version for colours)



It is possible to perceive the formation of images known in the literature, which allows different interpretations. In the co-citation analysis, it is possible to identify a 'pivot' type figure. In this type of figure, finding nodes between different networks is possible. This geometry type allows us to find publications linking different research groups and themes.

In the figure of the co-authorship analysis, a central node can be perceived peripheral nodes connect that. This geometry is known as a 'hub'. This type of figure allows us to interpret that the central node is a strong candidate for significant contributions to the literature analysed (Chen, 2004).

5 Discussion

Analysing Table 3, we noticed that the literature still needs answers regarding project context related to project management methodologies, techniques, and other management skills that are core for project management, such as stakeholder and risk management.

Studies mention that the project management approach exists, but achieving clear and precise results in project management environments and their specificities is necessary.

Thus, several issues are raised, highlighting the need to connect the usage of project management in agribusiness operations with performance, risk management, stakeholder management, and the business ecosystem.

We identified that the research on the relationship between performance and project management is still in an embryonic phase for this market, and there is a need for more studies addressing how the lack of methodologies in project management affects their success.

Even in successful cases with significant practical and theoretical implications, as in the study proposed by Zucchella and Previtali (2018), there needs to be a reference to using project management practices, methodologies, and tools. Nevertheless, the article addresses issues such as developing new business models in an innovation ecosystem.

Table 4 Open questions

<i>Topic area</i>		<i>Key findings</i>	<i>Unsolved questions</i>	<i>References</i>
Project management in agribusiness supply chain	1	Identified project management as an enabler to achieve results, increase productivity, efficiency and as a way to simulate and modulate results	How was the project management team organised? What methodologies from communities of practice were used?	Grammig (2003), Carlsson and Rönnqvist (2005), Rao (2007), Lejars et al. (2008), Essien et al. (2018), Zhang et al. (2016) and Busato et al. (2019)
	2	Stakeholder management as critical success factor	How are stakeholder management techniques used to improve project results?	Candelo et al. (2018)
	3	Risk management	How are risks managed when a new product, process or result is proposed?	Sallabi et al. (2011), Nyamah et al. (2017) and Óskarsdóttir and Oddsson (2018)
	4	Business ecosystem	How can project management support development of business ecosystem?	Homrich et al. (2018), Alfaro and Miller (2014) and Anbarasan and Sushil (2017)

The limitations found in this article result from the researchers' methodological choices (Wacker, 2004). Bibliometric analysis may have deviations or trends since they are performed based on the number of citations of each article, with the most cited articles generating temporal trends. Therefore, content analysis and snowball techniques were applied as mitigation tools.

When we interpret the analysis of networks, we perceive a structure (co-citation, keywords, and co-authorship) with a low number of diffuse relationships, showing a low degree of intersection and correlation between the terms 'project management, agriculture, and supply chain.'

The same interpretation can be made regarding bibliometric analysis; despite having journals relevant to academia with a high degree of citations, it was not possible to perceive a positive dynamism of publications; that is, there are many journals with a few articles each.

The content analysis noticed a prevalence trend in simulation studies and conceptual, theoretical models without financial support. In a qualitative approach, a period of retrospective analysis was carried out with the support of universities.

A highlight of the content analysis concerns the presence of project management methodology, project office structure, and the presence of project manager.

The content analysis showed that none of the analysed articles mentioned these structures. It was also not possible to identify links between project management practices considered by the academy and practitioners as an international reference in the research carried out.

With this in mind, we responded to QP proposed by this study. Project management methodologies need to be further discussed by academics and practitioners in the context of the agricultural supply chain. Understanding which methodologies and how they were applied can support understanding project performance and results.

In addition, supply management in the agricultural chain can easily be identified as a high-risk practice. There needs to be a more frequent practice of methodology to identify, withdraw and assume risks (Nyamah et al., 2017).

In this context, the project manager's use of project methodology, and the consequent figure, act to eliminate the threat or protect the project against its impact. Hence, this would change the project management plan to eliminate the threat (Carvalho et al., 2018).

Considering the complexity of the agricultural supply chain and all the knowledge associated with sustainable management, implementing project offices are best practice for knowledge management and, consequently, its integration from robust sharing systems (Enberg, 2012).

The development of agriculture goes beyond increasing earnings and ensuring the sustainability of agricultural production but plays an essential role in developing countries' economies (Rao, 2007). It has been used as a strategic tool to achieve organisational goals, so knowledge acquisition through dynamic competencies is evidenced.

Thus, it would be helpful to suggest an expansion of the debate on the role of the project manager in publications that deal with the theme of project management in the agricultural supply chain as a form of theoretical and practical contribution. From this literature review, it is impossible to precisely conclude the role of the project manager as an integrating agent of knowledge in the agricultural supply chain.

A vast literature demonstrates that investing in developing project management skills and in the project team itself increases project performance. More than that, the project's success depends on maturity and competence (Carvalho et al., 2013). Notwithstanding, it is essential to consider it as a theoretical contribution to the discussion of traditional models of supply chain operation, which must undergo revisions and changes to meet current demands. We proposed that the project manager, and the project office, act as a factor of integration and acquisition of competencies (dynamic or not) from the receipt of strategic definitions, innovation, and the organisation's objectives and apply the methodologies ensuring performance and adaptability, always considering the context of complexity, dysfunctions, contingency, and risk.

COVID-19 global pandemic has amplified the pressure on the agriculture sector, inciting the need for sustainable agriculture more than ever. The role of project management is an excellent confederate for sustainable and precision agriculture. The increase in skilled labour in the form of specialists in the agricultural supply chain will

generate strategic skills for organisations that will be able to facilitate the gain of results and establish values aligned with the objectives of the organisation (Brones et al., 2014).

6 Conclusions

The agricultural market has a complex operation involving large undertakings such as road construction, warehouses, and soil preparation. In addition, new technologies such as the internet of things, digitisation, and georeferencing have strongly influenced it to gain efficiencies in recent years.

These characteristics make the need to use projects to obtain results almost imperative. However, not only that, in recent years, there has been intense pressure for agricultural results to be increasingly sustainable.

This paper reveals that project management practices have yet to be used in the agricultural supply chain since there is no mention of the role of the project manager and rare mentions of project management techniques.

This brings us to the contributions of this paper. From a theoretical point of view, we have the main contributions:

- 1 it highlights opportunities for the use of project management methodologies and practices in the agricultural market as a practice of knowledge integration and dynamic acquisition of skills
- 2 it proposes the correlation between different stakeholders and strategic demands of companies as a way to guide professionals and academics to discuss the absence of these structures
- 3 can list the literature review regarding project management practices in the agricultural production chain.

From the point of view of managerial contribution, this paper highlights the power that the figure of the PM and the structure of the PMO have on the success of the project, indicating an opportunity to improve the results if this practice is adopted and the opportunity to manage the agricultural business with global sustainable development.

In future research, case studies are suggested to understand the factors, restrictions, or limitations imposed by the agricultural production chain affected by project management methodologies, the project manager, and the consequent presence of a project office.

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